CS536 Bonus Project: Developing Smart Glass Application (optional)

Submission Due: 23:59:59 PM, Monday Nov 30, 2026
In-class Presentation and Demo: Friday Dec 4, 2026
Total points: up to 240 points

CS592-DRN @Purdue (Chunyi Peng)

1 Goals and Timeline

This project is to develop one application on smart glasses and assess its performance in the wild. This project consists of the following parts:

- **Part A (50 point):** develop a live steaming application on smart glass + Android smartphone (+ laptop). *Demo in class required.*
- **Part B (50 points):** develop one AI application (out of multiple options) over Part A. *Demo in class required.*
- **Part C (50 points):** Performance (e.g., latency, stalls/failure) assessment in the wild.
- **Part D (50 points):** Enhancements over Part A-C and preliminary evaluation in the wild.
- **Part E (40 points):** A technical report (4-6 pages, 10pt, double column), presentation and demo in class.

Note: this project is optional. The total points are equal to the sum of all the labs but the required efforts are larger than the total workload of all the programming labs. Please think thoroughly and decide whether to do this bonus project.

If you plan to do this bonus project, you need to complete all the following tasks.

- **Build your team**. Working alone or with a team up to four students is allowed. Note that all the points earned in this bonus project will be equally divided among all the team members. For example, each member earns 110 points if the whole team of two members gets 220 points out of 240 points.
- **Registration at Gradescope (Sep 25 - Oct 16)**. Any team must submit one page including all the team members and the parts planned to do. Note that you can choose to do some but not all the parts required by this bonus project. Changes throughout this project period are allowed and understood.

  If you have any questions before the registration, please contact us (cs536-ta-fall26@cs.purdue.edu) with the subject that starts with **"CS536: BONUS"** or talk to Prof Peng before/after the class. (No Q&A at Ed Discussion).

- **Mid-term check at Gradescope (Nov 16)**. The participating team must submit one page that summarizes what you have done so far and the parts that you plan to complete for this bonus project. Note that the team must clearly specify the changes (those parts different from those planned at the registration).

  If any team wants to quit, please contact us (cs536-ta-fall26@cs.purdue.edu) with the subject that starts with **"CS536: BONUS QUIT"** or talk to Prof Peng before/after the class. (No Q&A at Ed Discussion).

- **Submission at Gradescope (Nov 30)**. The participating team must submit one page that summarizes what you have done for this project, and what will be covered in the in-class demo and presentation.

- **In-class Demo and Presentation (Dec 4)** . Every team must do in-class presentation and demo. A full technical report (4-6 pages) should be submitted before or on Dec 4 (No penalty if the report is received before Dec 6 Midnight).

  Note: more guidelines will follow when approaching to the end of this semester.

2 Part A: Develop a live streaming app for Smart Glass (50 points)

In this part, you need to develop a live streaming application for one pair of smart glasses. You will use one glass model: IMNO Air 2 or IMNO Air 3, which is available by contacting Prof Peng. Please use Github (a private repo) to manage this project. Once completed, please save your codes at Github with one tag of "BONUS-PartA".

You need to connect the smart glasses with an Android smartphone and stream whatever is seen by the smart glasses into one Android application. For the sake of easier development and debugging, please connect your smartphone with one laptop so that all the streamed content can be displayed on the laptop. Please make sure that the developed streaming application works on the Smartglass + Android Smartphone + Laptop environment. Such demo in class is required. In addition, you will need to submit your code (using Github) as well.

Tip: please test the developed application by live streaming for more than 1 minute before you move to Part B.

3 Part B: Develop an AI application over Smart Glass (50 points)

Please develop one of the following AI application over Part A:

- Google Lens or alike. Please search whatever your glasses see and display the results on the glasses in a user-friendly way.

- Google translate or alike. Please translate whatever texts you see from one language to another language. Note that your application must decently handle if no texts are seen.

- One computer vision task, including but not limited to gesture recognition, object detection and tracking (say, face or human detection and tracking), semantic segmentation, 3D scene reconstruction.

- DIY. Defined by your interest which leverages recent AI techniques. Need to contact Prof Peng to get her approval in advance.

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Note that your application can be configured in advance (settings) and controlled at runtime (start, stop and pause). Once completed, please save your codes at Github with one tag of "BONUS-PartB". You need to submit your code (using Github). Moreover, the demo in class is also required.

4 Part C: Performance Evaluation in the Wild (50 points)

Please run the field trails to evaluate the performance in the wild.

- Please consider multiple use scenarios in the lab/office/campus and home etc (over different wireless networks: PAL3.0 and 5G). For each use scenario, please ensure 10+ runs (each run *ge* 30 seconds, typically 1-5 min). In total, the experiment time should be more than 1 hour.
- Please evaluate the performance of streaming (Part A) using the following metrics on networking performance: throughput (uplink and downlink), latency, failure etc and streaming-specific KPIs (stalls, jitters etc).
- Regarding the AI application (Part B), please evaluate the user experience with app-specific KPIs. (hint: you need to define KPIs based on the developed AI app).
- When you run every filed experiment, please save all the raw traces, which will be used to analyze network performance.

Please put your scripts that process traces into a new folder called "Bonus-PartC". You need to analyze all these traces and plot the results in the figures. The results in figures will be needed in your presentation and final report. Once complete, please save your codes at Github with one tag of "BONUS-PartC".

Note: you need to store all the raw logs, which are not required for submission but might be requested on demand.

5 Part D: Enhancements and Preliminary Evaluation (50 points)

Based on the results in Part C, please enhance one aspect of the developed system to improve user experience or network performance.

Please describe the proposed solution for the identified problem in Part C. Please implement the proposed enhancement and evaluate it in the wild (note that that scale can be smaller as the preliminary evaluation is required).

In your presentation and final report, please describe your efforts and results. All the new codes will be marked with the last tag of "BONUS-PartD".

6 Materials to turn in

- **Codes**: You will submit your codes at **Gradescope**. Please submit your code by uploading your github URL (if it is a private repo, please add "chunyip" into the collaborator, or make it a public repo till the end of grading window). Otherwise, you need to organize all the codes and zip them along with README file. You need to submit Bonus-PartA.zip, Bonus-PartB.zip, Bonus-PartC.zip and Bonus-PartD.zip if applicable.
- **In-class presentation and Demo**: A 10-min presentation in class is required. In the meanwhile, demos for Part A and Part B are required. Demo for Part D is encouraged but optional.

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- **Final Report (pdf)**: Please use Latex to complete your final report (4-6 pages, 10pt, two column). Please use the latest Latex template (Version 2.20): https://www.acm.org/publications/proceedings-template Overleaf is another good choice for you with the latex template available. Please use [sigconf,10pt]acmart

  Please describe what you have done in this bonus project, as well as the results (the performance evaluation and enhancements must be presented in texts and figures).

7 Support

Questions about this bonus project should be emailed to `cs536-ta-fall26@cs.purdue.edu` or `chunyi@purdue.edu` with the subject that starts with **"CS536: BONUS"** or asked before/after the class or Prof Peng's office hour (No Q&A at Ed Discussion).

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